

3 | Research Article **OPEN ACCESS**4 **A real-time reservoir inflow forecast evaluation**
5 **framework**

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7 ¹Pacific Northwest National Laboratory, Washington, USA |8 **Correspondence:** Corresponding author Cameron Bracken, cameron.bracken@pnnl.gov9 **Received:** Date Month Year | **Revised:** Date Month Year | **Accepted:** Date Month Year10 **Keywords:** hydropower, streamflow forecasting, real-time evaluation, verification, validation**Abstract**

Recent advances in hydrologic forecasting have produced a growing number of streamflow forecast products. Hydropower operators must determine whether these products improve operations enough to justify their cost. We present a framework for evaluating competing reservoir inflow forecasts under current operating conditions. Operators participate in selecting and revising the metrics, and the evaluation uses the same data available in daily operations. This approach combines standard verification metrics with measures tailored to site-specific decisions. We apply the framework to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U.S. The commercial product labeled B generally matched the in-house forecast and outperformed product A. Under an idealized cumulative-error stress test, product B also had a lower probability of exceeding the forebay compliance band during the first 24 hours at all three dams. Its advantage persisted through the 32-hour horizon shared by all products, although exceedance probabilities were high enough that none of the forecasts could be used without operator intervention. The framework evaluates forecast quality rather than economic value. A value assessment would also require a scheduling model, operating constraints, and market prices.

11

12 **1 | Introduction**

Changes in environmental regulation and water demand, along with growing operational uncertainty, require more precise operation of reservoir and hydropower systems (e.g., [Lee et al. \(2025\)](#) and [Guo and Liu \(2024\)](#) and [Badr et al. \(2023\)](#) and [Aljoda and Jain \(2021\)](#)). These demands have increased interest in streamflow forecasting tools ([Yang et al. 2021](#) and [Pagano et al. 2014](#) and [Liu et al. 2012](#)). With numerous commercial and in-house products available, water managers must determine whether a new product will improve their operations enough to justify the associated cost.

Hydrologic forecasting and hydropower scheduling have improved through data driven models ([Kratzert et al. 2019](#)) and scheduling optimization ([Zhang et al. 2024](#)). Forecast verification, however, still commonly relies on hindcasts evaluated with standard statistical metrics over a long, representative period. Hindcasts may be unavailable because they are costly to produce or require data that were not archived. Archived operational forecasts offer an alternative, but they fit poorly when products change during the evaluation or are adjusted by operators. This study addresses the case in which reservoir operators must compare products under current operating conditions without a hindcast.

Pairing a forecast product with the needs of a particular operator is complicated by differences in meteorological and hydrologic data, watershed properties, and infrastructure among basins ([Pham et al. 2021](#)). Operating rules range from run-of-river constraints to seasonal rule curves ([Zajac et al. 2017](#)), and the relevant decisions change with forecast lead

time and flow regime. An evaluation must account for these differences and connect statistical performance to decisions at a particular site.

Several programs have compared forecasts under common conditions. HEPEX has organized community experiments and testbeds¹, and the U.S. Bureau of Reclamation has conducted forecast rodeos². The hydropower industry often favors evaluations that fit within current operations (Krajewski et al. 2021). These comparisons provide useful information about forecast performance, but a common ranking does not show whether a product will improve decisions at a particular utility enough to cover deployment, training, and transition costs.

We present a framework for evaluating streamflow forecasts in this setting. Operators participate in the experimental design, select and revise metrics, and interpret the results in relation to recent decisions. The evaluation uses the data available in daily operations. Operator participation and the ability to revise metrics as data are collected distinguish the framework from a fixed retrospective verification.

We demonstrate the framework at three Great River Hydro plants on the Connecticut River. The case study combines standard metrics with measures tied to day-ahead bidding and forebay compliance. Although the tailored metrics will differ among systems, the evaluation process can be adapted to other river basins and operating settings.

2 | Methods

Although probabilistic and ensemble forecasts are increasingly available, deterministic forecasts such as those issued by the National Weather Service River Forecast Centers (RFCs) remain widely used in operations. All three products evaluated here are deterministic, so we focus on deterministic metrics that can be computed in real time. Section 2.5 describes how the same process can be applied to ensemble products.

Hindcasts provide the strongest statistical basis for evaluating forecast quality because they cover large samples and a broad range of hydrologic conditions. Comparing hindcast inflow with observed or calculated inflow does not require reconstructing reservoir operations. Forecasts that incorporate human judgment are more difficult to reconstruct because the forecaster would need the information available at each historical issue time. Historical wholesale prices can often be retrieved, but outages, operating constraints, operator judgment, and the influence of price signals on decisions may not be recoverable. Extending an evaluation from forecast quality to forecast value also requires a reservoir operations model or operator input. A real-time evaluation develops the forecast archive while these conditions occur and records the decisions made in response to them.

2.1 | Real-time inflow forecast evaluation

A general real-time forecast evaluation process can be described in the following steps. Steps 1 through 3 are generic practice, common to any verification exercise. Steps 4 through 6 are where this framework departs from retrospective verification.

1. **Collect real-time data** - This may include streamflow forecasts (natural or regulated), calculated inflows, scheduled and actual outflows, and other data relevant for computing evaluation-relevant metrics such as forecast issue times and past forecasts.
2. **Compute evaluation metrics** - At the start of an evaluation, it may only be possible to compute a limited number of generic evaluation metrics but as the available data grows it may become possible to compute more reliable metrics as well as metrics tailored to the specific system.
3. **Visualize results** - Develop clear figures that allow operators to compare products and identify differences that matter to operations.
4. **Gather feedback** - Collecting feedback on an ongoing basis from operators is critical to the success of a real-time forecast evaluation. Operators should be asked which recent forecast errors mattered operationally and which did not, since the answers are not always apparent from the metrics alone.
5. **Update metrics** - Feedback is used to add metrics tailored to the system or reservoir being evaluated, and to remove or modify metrics that have proven uninformative. The set of metrics is therefore not fixed at the outset but develops as the evaluation proceeds.
6. **Iterate** - The above steps should be repeated at a regular interval (monthly or weekly) throughout the entire forecast evaluation period, ideally at least one year in duration.

¹ www.hepex.org

² <https://www.usbr.gov/research/challenges/forecastrodeo.html>

The mechanics of steps 4 and 5 determine what the evaluation produces, so an example of how feedback was collected and acted on in our case helps clarify this point. Feedback was collected in recurring meetings with the operators, held monthly for most of the evaluation. Each meeting reviewed the current calculated metrics against recent operating experience, and operators were asked to identify which recent forecast errors mattered operationally, and which did not. The answers frequently diverged from what the generic metrics indicated, especially during the high-flow season. Operators discounted large errors during high-flow spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to impact the forebay compliance band as serious. These ongoing discussions motivated the tailored metrics in Section 3.6: the forebay exceedance metrics were designed in direct response to operators stating that compliance risk, rather than average error, determined whether they trusted a forecast.

The iteration in step 6 occurs over two timescales. In this case, the set of metrics stabilized within about six months, while the evaluation continued to collect data under a wider range of conditions. For applications with strong seasonality, including this one, we recommend at least one annual cycle because product rankings can vary by season (Section 3.5). Operators should expect the metrics to stabilize before the evaluation period ends. Conclusions from a shorter evaluation remain conditional on the seasons and events observed.

2.2 | Benchmark forecasts

Benchmark forecasts put product performance in context by showing how much improvement a product provides over a known alternative. A “perfect” forecast substitutes observed or calculated inflow for a forecast and represents the upper reference. A persistence forecast carries the most recent observation through the forecast horizon. Persistence is simple but can be competitive from hourly lead times through about two days, making it an appropriate reference for day-ahead operations. It degrades quickly beyond that horizon, where climatology is usually a more suitable lower reference. Evaluators should match the benchmark to the decision horizon rather than apply one reference at all lead times. Other useful benchmarks include public forecast products and the forecast currently used by the reservoir operator.

Benchmarks are also useful when comparing results among locations. NSE and KGE depend on the distribution of observations at each location, so their raw values are not directly comparable when inflow variability differs (Knoben et al. 2019). NSE, for example, measures improvement over the observed mean. The same absolute forecast error can produce different NSE values at two sites because the observed variances differ. A forecast can therefore have a higher NSE at a more variable site even when its absolute errors are larger. We report RMSE and reference forecasts alongside NSE and KGE so that the results show both absolute error and performance relative to alternatives available to the operator.

2.3 | Smoothing calculated inflows

Reservoir inflow is often calculated from changes in forebay elevation and an elevation-volume relationship rather than measured directly. Short-term changes caused by wind, turbine and spillway releases, seiches, and waves from upstream can introduce noise and nonphysical values into the calculated inflow. Smoothing may therefore be appropriate before calculated and forecast inflow are compared. We used two successive moving averages, which produce a triangular weighting function, but other filters are available (Elshorbagy et al. 2002). The smoothing window should correspond to the timescale of the decision. A window that is too short will retain noise, while one that is too long may remove variability that the forecast should reproduce. Because smoothing changes the quantity being evaluated, the filter should be reported and the sensitivity of the results to its width should be tested.

2.4 | Traditional inflow evaluation metrics

Deterministic inflow forecasts can be evaluated using measures of average error, correlation, and bias. Higher moments, categorical performance, and conditional statistics may also be useful (St-Aubin and Agard 2022 and Hewamalage et al. 2023). We do not attempt a complete review of deterministic forecast metrics. Instead, we use a small set that describes several aspects of performance and can be expanded for a particular application. No single metric fully describes forecast quality. The metrics should also be calculated by lead time and season because values aggregated across the full record can hide changes that matter to operations.

Average hydrologic forecast behavior is typically measured with an error metric such as root mean squared error (RMSE), Nash-Sutcliffe Efficiency (NSE) (Nash and Sutcliffe 1970), or Kling-Gupta Efficiency (KGE) (Gupta et al. 2009). We recommend including KGE because it decomposes forecast performance into correlation, a variability ratio, and a bias ratio.

A low score can therefore be associated with a specific deficiency rather than registered only as a large aggregate error. NSE and RMSE aggregate squared error into a single number, which makes them more sensitive to the largest errors and less diagnostic about their source. The metrics answer different questions, so we report them together rather than use one as a complete measure of forecast quality. The KGE is given by

$$\text{KGE} = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_f}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_f}{\mu_o} - 1\right)^2} \quad (1)$$

where r is the linear correlation coefficient, σ_f and σ_o are the forecast and observed standard deviations, and μ_f and μ_o are the forecast and observed means. KGE values greater than -0.41 improve on the observed mean as a predictor (Knoben et al. 2019), and the maximum value is 1 for a perfect forecast. Because the corresponding reference value for NSE is 0, the two efficiency metrics should not be interpreted on the same scale. We report both and use RMSE, bias, correlation, and benchmark forecasts to interpret their values.

The error, or residual, of a single forecast is the difference between the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts as a percentage of total observed volume

$$\text{PBIAS} = 100 [\%] \times \frac{\sum(f_i - o_i)}{\sum(o_i)} \quad (2)$$

where positive values indicate forecasts greater than observations and negative values indicate forecasts less than observations.

Linear association between observations and forecasts is measured using the Pearson correlation coefficient

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2 \sum(f_i - \mu_f)^2}} \quad (3)$$

where r can range from -1 to 1 but for forecast evaluation, negative values would indicate poor forecast performance.

Metrics should be computed on a common sample when products cover different periods. Skill scores relative to a reference forecast are also useful, but the evaluated and reference forecasts must use the same observations. Otherwise differences in sample variance can be mistaken for differences in skill.

2.5 | Extension to probabilistic and ensemble forecasts

The process described in this section applies to both deterministic and probabilistic forecasts; only the metrics and visualizations change. Probabilistic products can describe uncertainty around compliance or market decisions rather than provide only a single expected value. Where ensemble or probabilistic products are available, the continuous ranked probability score (CRPS) can summarize the accuracy of the full predictive distribution. CRPS reduces to mean absolute error for a deterministic forecast, which permits comparison on a common scale. Rank histograms can indicate whether an ensemble is biased or has too little or too much spread, while reliability diagrams compare forecast probabilities with observed frequencies. The choice among these metrics should follow the same process used for deterministic forecasts and reflect the decisions being evaluated.

3 | Case Study: Great River Hydro

Great River Hydro (GRH) is New England's largest conventional hydropower generator. As a private, independent power producer, GRH owns and operates 589 megawatts of installed nameplate capacity across 13 hydropower plants on the Connecticut and Deerfield Rivers in ISO New England. We evaluated inflow forecasts for three mainstem dams, Wilder, Bellows Falls, and Vernon (Figure 1), and for three tributaries, the Ottauquechee, Sugar, and White Rivers.

The three dams lie in series on the main stem of the Connecticut River, with Wilder upstream, followed by Bellows Falls and Vernon. They are operated jointly, and releases from each dam become inflow to the next. The three tributaries listed above provide the main local inflows between Wilder and Bellows Falls. We selected these sites from the larger set shown in Figure 1 because they are the largest generators in the system and receive most of the operators' scheduling attention. The projects are also part of an ongoing relicensing process through the Federal Energy Regulatory Commission (FERC) that includes new limits on forebay fluctuations. The remaining facilities in Figure 1 are smaller and are not covered by the same license conditions.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers, which contributes to inflow variability that is not directly weather driven.

TABLE 1 | Physical characteristics of the six evaluation points. Drainage areas are from the USGS National Water Information System for the reference gage listed. Mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder.

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawaquechee River	Tributary	01151500 North Hartland, VT	221	572

3.1 | Operating context

The three dams have limited storage relative to their inflow and are operated as run-of-river facilities. Their daily cycle consists of an upward ramp, a period of flexible generation, a downward ramp, and a refill period. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most to the operators, since it sets what they can bid into the market. The binding operational constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation, with penalties for exceedance. The central operating problem is therefore committing generation based on an uncertain inflow forecast without violating the forebay limits. This is what motivated the cumulative-error and threshold-exceedance metrics in Section 3.6.

At the time of the evaluation the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent product against which the two commercial forecasts were evaluated, and it is labeled GRH in the results.

Setting up a real-time evaluation required the development of an automated data platform that enabled sharing of real-time system data as well as the forecasts. This was necessary to share certain data with external parties for the purposes of forecast evaluation. While such a system may not be necessary for evaluations conducted internally by a single organization, some form of data access (API, database, etc.) is required for a real-time forecast evaluation. Such systems incur an up-front cost that should be considered when deciding on an evaluation approach. In addition, real-time evaluations come with all of the challenges of any operational system, such as system uptime, latency, and maintenance. These factors may help decide if a forecast is viable before any evaluation is done; a forecast that cannot be produced operationally should not be considered in a real-time evaluation.

We conducted the real-time evaluation for almost two years, from June 2023 through April 2025, with regular feedback from the operators. We evaluated two commercial forecast products, anonymized as A and B, and the GRH in-house forecast, which combines RFC forecasts, routing, and operator judgment. During the first few months, we focused on general measures of forecast performance. As the archive grew, we added seasonal summaries and metrics developed with the system operators.

The products cover different portions of this period, which limits direct comparison. Forecast B was available for the full period, providing 657 day-ahead forecasts at each dam. Forecast A was discontinued in November 2024, and the in-house forecast archive also ends in November 2024, providing 539 and 528 day-ahead forecasts, respectively. The three products overlap on 510 days at each dam, from 3 June 2023 through 20 November 2024. We report metrics computed from each product’s full record, so the samples are not identical. We repeated the calculations using the 510 common days. The product rankings were unchanged at all three dams, and individual NSE values changed by no more than 0.03. We therefore retained the full record for each product. Partial coverage of this kind is a practical limitation of real-time evaluations.

3.2 | Forecast products

Anonymization of the commercial products limits reproducibility, so we describe each to the extent our agreements allow. Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the

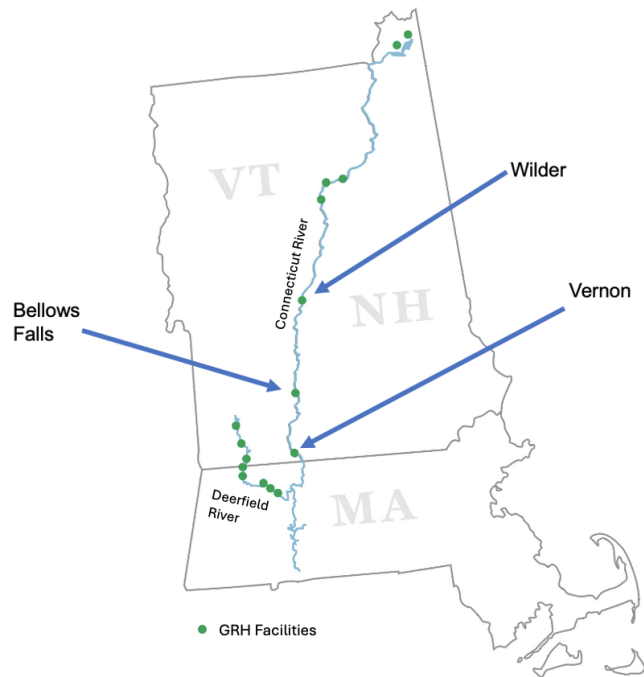


FIGURE 1 | The Great River Hydro system in the Northeast U.S. with the locations of the three major dams in this evaluation.

Connecticut River main stem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly time step out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at an hourly time step out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework. A full procurement evaluation should also weigh the operational cost, data dependencies, and ease of integration of each product alongside its skill, since these differ substantially between the two commercial products evaluated here.

3.3 | Hydrology of the evaluation period

The evaluation period included the July 2023 Vermont flood. This event produced the largest calculated inflows of the evaluation at Vernon (83,100 cfs, 2,353 m³/s) and Bellows Falls (67,000 cfs). A December 2023 rain-on-snow event produced the largest inflow at Wilder (27,600 cfs) and the second largest at Vernon and Bellows Falls. The evaluation period also included spring freshets of 2024 and 2025 and dry summer conditions in both 2023 and 2024. The period therefore sampled a reasonable range of the conditions relevant to GRH operations. It did not, however, sample the full distribution of hydrologic conditions. Notably, the July 2023 event produced one of the ten largest flows ever recorded on the main stem of the Connecticut River. The evaluation also did not include any severe multi-year drought. Any real-time evaluation is bounded in this way, and the conclusions below should be taken as conditional on the observed period. Most weather driven inflow forecasts extend 7–10 days, but GRH relies most on next-day volume because it informs the day-ahead market bid. Table 2 compares the two commercial products, the in-house forecast, and the perfect and persistence benchmarks at this horizon. Over the evaluation period, forecast B and the in-house forecast performed similarly, and both outperformed A. The lead-time and monthly results below show variation that is hidden by the aggregate values. Table 2 also shows why cross-location comparisons need more than one metric. Forecast A has its highest NSE at Vernon (0.81) and its lowest at Bellows Falls (0.55), yet its RMSE is 5,097 cfs at Vernon, 5,904 cfs at Bellows Falls, and 2,278 cfs at Wilder. NSE is normalized by observed variance, so a higher value at one site does not imply a smaller error than at another. The persistence benchmark also changes sharply among sites, from an NSE of −0.81 at Wilder to 0.80 at Vernon. We therefore use the table to compare products within a location, while RMSE and the benchmark rows provide context across locations. Metrics computed on the 510 days common to all products preserve the within-location ranking.

TABLE 2 | Day-ahead forecast evaluation metrics for June 2023 to April 2025. A and B are commercial products, GRH is the in-house forecast, and perfect and persistence are benchmarks. The products cover different portions of the period; n is the number of forecasts evaluated. Values should be compared primarily within a location because NSE and KGE depend on the local observation distribution.

Location	Forecast	n	NSE	KGE	PBIAS (%)	RMSE (cfs)	r
Wilder	A	539	0.72	0.83	−8.0	2,278	0.88
	B	657	0.83	0.88	0.1	1,786	0.93
	GRH	528	0.88	0.89	−9.9	1,476	0.95
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	−0.81	0.02	−75.9	5,866	0.75
Bellows Falls	A	539	0.55	0.77	0.4	5,904	0.77
	B	657	0.91	0.95	0.6	2,615	0.95
	GRH	528	0.85	0.88	−10.4	3,309	0.93
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	0.39	0.35	−39.2	6,831	0.87
Vernon	A	539	0.81	0.83	−11.3	5,097	0.91
	B	657	0.89	0.94	1.7	3,872	0.94
	GRH	528	0.89	0.88	−11.0	3,764	0.95
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	0.80	0.69	−18.1	5,167	0.93

Figure 2 shows forecast and benchmark performance through a 10-day lead time. The GRH in-house forecast is used for day-ahead market bids and does not extend through the full period. At Wilder and Vernon, the three forecasts perform similarly over the first 1–3 days, after which A and B diverge. The differences are larger at Bellows Falls, where B performs better at most lead times, followed by the GRH forecast and then A. Both commercial forecasts develop an increasingly negative bias with lead time, and this bias increases downstream, consistent with errors propagating through the cascade. At Bellows Falls, for example, the PBIAS for A changes from 0.4% at the day-ahead horizon to −19.1% by day 7. The corresponding values for B are −1.3% and −7.8%. Because releases from each dam become inflow to the next, forecast errors can propagate to the next downstream project and accumulate with lead time.

3.4 | Smoothing GRH calculated inflows

Figure 3 shows calculated inflow before and after smoothing. We applied two successive 24-hour moving averages to the hourly calculated inflow. Their convolution yields a triangular kernel spanning 47 hourly observations; “24-hour” refers to the width of each component moving average. We selected this width to match the day-ahead volumetric decision and remove fluctuations at shorter timescales. Because the window directly affects the metrics, we repeated the day-ahead evaluation with component widths of 0, 6, 12, and 24 hours (Table 3). The metrics improve slightly as the width increases, as expected when observation noise is removed. The largest change is 0.05 NSE. The ranking of the products is unchanged at every width and location. At Bellows Falls, for example, B leads A by 0.39 NSE without smoothing and by 0.37 with the 24-hour component width. The conclusions therefore do not depend on the selected window. Evaluators should still report the filter and confirm that their results are insensitive to its width because excessive smoothing could favor forecasts that underrepresent inflow variability.

TABLE 3 | Sensitivity of day-ahead NSE to the component moving-average width used to smooth calculated inflow. Each nonzero width is applied twice, producing a triangular kernel. The 24-hour column corresponds to Table 2. Product rankings are unchanged.

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

249 **3.5 | Seasonal performance**

250 Figure 4 shows monthly variation in forecast quality. All three products have their highest skill during the spring snowmelt
251 period, roughly March through May. The commercial products have their lowest skill in late summer and fall. Forecast
252 A also degrades in January and February, when its day-ahead KGE pooled across the three dams falls to 0.45 and −0.67,
253 respectively; the February value is below the plotted range. Forecast performance is lower when inflow depends on rain,
254 rain-on-snow, or convective storms, all of which depend on uncertain precipitation forecasts. Snowmelt inflow is more
255 predictable because the snowpack can be observed before it melts and temperature forecasts are generally more reliable
256 than precipitation forecasts. Forecast A outperforms B in April and May, so no product leads in every month.

257 **3.6 | Tailored metrics**

258 In addition to the general evaluation metrics described above, we developed several metrics for the GRH system and its
259 operations. Each arose from the feedback process described in steps 4 and 5 of Section 2.1. Table 4 traces each metric to the
260 operator concern that prompted it. The first measures day-ahead performance because GRH uses the next day's expected
261 inflow to prepare its day-ahead market bid. Operators adjust flows daily, so they concentrate on the first two days and do
262 not use a forecast beyond that horizon.

TABLE 4 | Tailored metrics developed during the evaluation, the operator concern that prompted each, and the approximate point in the evaluation at which it was added. Metric development converged within roughly the first six months.

Metric	Operator concern	Added
Day-ahead volumetric skill	Day-ahead market bids are set from next-day inflow volume, not hourly shape	Months 1–2
Seasonal metric breakdown	Products were suspected to behave differently in snowmelt and rain driven regimes	Months 3–4
Cumulative forebay error	Average error does not indicate whether the forebay band would be breached	Months 4–6
Forebay exceedance probability	Operators needed to know how quickly cumulative forecast error could exhaust the forebay band	Months 5–6

263 The FERC relicensing described in Section 3.1 imposed a ± 0.5 foot (≈ 15 cm) forebay operating band. We calculated cu-
264 mulative hourly forecast error within each forecast issue and converted the accumulated volume to an equivalent forebay
265 elevation change using an approximate storage scale of 30,000 cfs-hours per foot. Figure 5 shows the median and middle
266 95% of the resulting trajectories. This is an idealized stress test: it assumes forecast inflow is passed directly as outflow
267 from a neutral forebay position. GRH does not operate this way. Operators schedule generation to market bids, depart
268 from inflow within the day, and adjust as new information arrives. The metric is therefore neither a prediction nor a
269 bound on observed compliance rates. Estimating realized risk requires a scheduling model and operator review.

270 Figure 6 shows the corresponding probability of leaving the forebay band. Over the first 24 hours of each product's horizon,
271 forecast B has lower exceedance probability than A at all three dams: 31.0% versus 56.6% at Wilder, 47.7% versus 67.9% at
272 Bellows Falls, and 53.8% versus 71.1% at Vernon. The in-house forecast, whose horizon extends only 32 hours past issue,
273 gives 53.9%, 67.6%, and 73.3% respectively, placing it close to A rather than to B. By 32 hours the ordering is unchanged:
274 B reaches 42.8–65.5% across the three dams while A reaches 61.2–82.0% and the in-house forecast 60.2–85.4%. Thus B
275 holds a consistent advantage across the whole window the three products share, but no product can be passed through
276 without intervention: by the end of the first day every one of them leaves the band in roughly a third to three-quarters of
277 issues, and the exceedance probability rises monotonically with each additional hour accumulated. This distinction was
278 not apparent from the aggregate metrics alone.

279 **4 | Discussion**

280 We present a framework for evaluating reservoir inflow forecasts in real time or near real time. It is intended for utilities
281 that need to compare several products under current operating conditions when a hindcast is unavailable. The framework
282 combines standard forecast verification with an iterative process in which operators identify the errors that matter to
283 their decisions and help revise the metrics. We applied this process to three reservoirs in the Great River Hydro system.
284 Product B generally matched the in-house forecast and outperformed product A. The forebay stress test also expressed

285 this difference in operational units and showed how the risk accumulated over the shared forecast horizon. These results
286 can help screen forecast products, but they do not by themselves estimate revenue or observed compliance outcomes.

287 **4.1 | Comparison with retrospective forecast evaluation**

288 Because these results were produced through an ongoing real-time evaluation, it is reasonable to ask what a retrospective
289 evaluation would add. Hindcasts can provide larger samples, broader hydrologic coverage, and more examples of extreme
290 events than the less than two years available here. They can be compared directly with observed or calculated inflow
291 to assess forecast quality. Reconstructing forecast value is more difficult. Historical wholesale prices may be available,
292 but outages, operating constraints, operator judgment, and the response to market conditions may not be. A real-time
293 evaluation records these conditions as they occur, but it does not replace a hindcast when one is feasible. We recommend
294 using both approaches when possible because they provide complementary information.

295 **4.2 | From forecast quality to forecast value**

296 The metrics presented here measure forecast quality rather than forecast value. Quality describes the relation between
297 forecasts and observations. Value also depends on the decision, its consequences, and the decision maker's tolerance for
298 risk. A more accurate forecast has no additional value if it does not change an action. Classical cost-loss analysis links
299 forecast use to the utility of a decision (Thompson and Brier 1955). The relative utility value framework extends this
300 approach to binary, multicategory, and continuous decisions and can represent risk aversion (Laugesen et al. 2023). Farkas
301 et al. (2025) review related methods.

302 The tailored metrics are a step toward value because they express forecast error in units related to a decision. The forebay
303 metric remains a stress test rather than a valuation because it does not include actual schedules or operator responses. A
304 value study at GRH could first screen products using the quality evaluation and then route the archived forecasts through
305 a scheduling and dispatch model. That model would need to represent reservoir constraints, unit performance, day-ahead
306 and real-time prices, and the current operating strategy. Benefits should be measured relative to the forecast and decisions
307 now in use and tested under different operating strategies and attitudes toward risk.

308 **4.3 | Limitations and next steps**

309 Forecast B performed better overall, but it did not lead in every month or at every forecast horizon. This variation illustrates
310 the difficulty of selecting one deterministic forecast for all conditions. Product selection should account for the decisions,
311 seasons, and horizons that matter most to the operator. Probabilistic forecasts can describe uncertainty around these
312 choices, although operators must still submit fixed market bids.

313 The results also show why forecast performance should not be compared among locations using raw efficiency metrics
314 alone. NSE and KGE depend on the distribution of observations, so a high score at one site does not necessarily imply
315 smaller errors than a lower score at another site. RMSE and reference forecasts provide additional context. Comparisons
316 among products should also use the same dates when their coverage differs. Here, restricting the analysis to the common
317 period did not change the product rankings.

318 We did not estimate how improvements in inflow forecasts affect hydropower operations or revenue. That analysis would
319 require coupling the archived forecasts to a hydropower scheduling model. A model configured to reproduce the current
320 bidding strategy may understate the benefit of a more accurate forecast because that strategy was developed around the
321 incumbent product. Estimating value may therefore require changes to the scheduling model and operating strategy as
322 well as the forecast input.

323 The length of the available record also limits the evaluation. The period included a regional flood and a rain-on-snow
324 event (Section 3.3), but it did not include a severe multi-year drought. We could not assess forecast performance under
325 sustained water-limited conditions. A multi-decade hindcast would include a wider range of hydrologic conditions and
326 provide a more complete evaluation of forecast quality.

327 The choice between real-time and retrospective evaluation depends on the needs of the system and the available resources.
328 We recommend conducting both when feasible. For a system with strong seasonal variation, a real-time evaluation should
329 span at least one year. A shorter period samples only part of the annual cycle, and its conclusions remain conditional on
330 the seasons and events observed.

331 Two other limitations affect the interpretation of the results. First, the reference inflows are calculated from forebay eleva-
332 tion changes and dam outflow. The observations therefore depend on reservoir operations and are not independent in the

333 same way as a stream gauge record. Second, the three products cover different portions of the evaluation period (Section
334 3.1), so the metrics in Table 2 use overlapping but nonidentical samples. Restricting the analysis to common dates does
335 not change the ranking. We report the full records to retain the available data.
336 The iterative component of the framework allows the evaluation to change as operators identify which forecast errors
337 affect their decisions. Standard metrics provide a consistent basis for comparison, while tailored metrics connect forecast
338 errors to the constraints of a particular system. Together, these metrics provide the forecast archive and initial assessment
339 needed for a later scheduling and value analysis. The same process can be applied elsewhere, but the relevant decisions
340 and tailored metrics must be defined with the operators of each system.

341 DATA AND CODE

342 Code for the evaluation is freely available³. Sample data can be accessed from the data repository on Zenodo⁴.
343 A preprint version of the manuscript is available at <https://eartharxiv.org/repository/view/10530/>.

344 AUTHOR CONTRIBUTIONS

345 Cameron Bracken - Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation,
346 Visualization, Writing - Original Draft Preparation. Youngjun Son - Conceptualization, Data Curation, Formal Analysis,
347 Investigation, Software, Validation, Visualization, Writing - Original Draft Preparation. Vince Tidwell - Conceptualization,
348 Formal Analysis, Investigation, Methodology, Project Administration, Supervision, Writing - Original Draft Preparation.
349 Nathalie Voisin - Conceptualization, Methodology, Project Administration, Supervision, Funding Acquisition, Writing -
350 Original Draft.

351 CONFLICT OF INTEREST

352 The authors declare no potential conflict of interests.

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³ <https://github.com/HydroWIREs-PNNL/inflow-forecast-evaluation>

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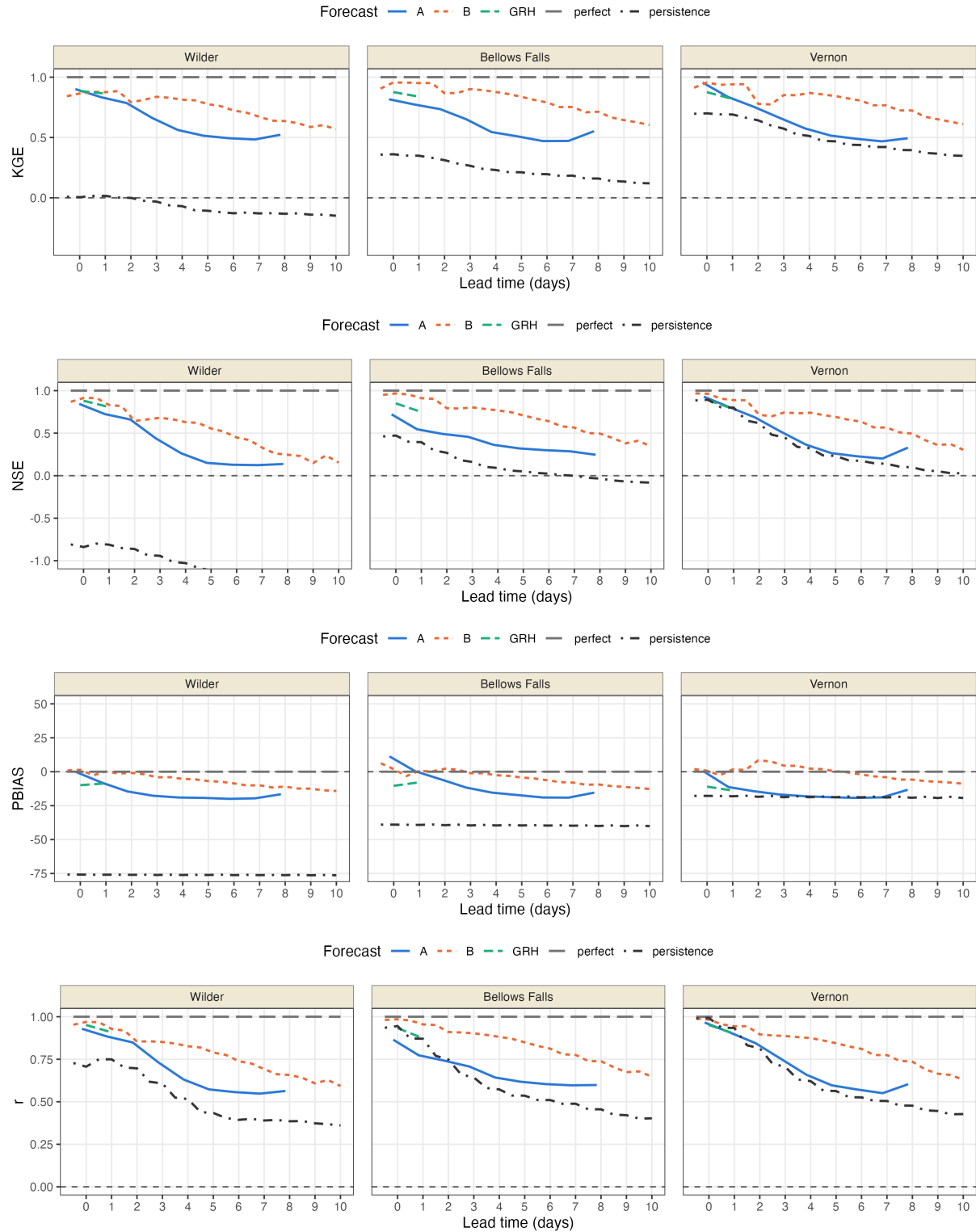


FIGURE 2 | Forecast performance metrics by lead time. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41 . The persistence benchmark is a meaningful reference only over the first two days; beyond that it is shown only for demonstration purposes.

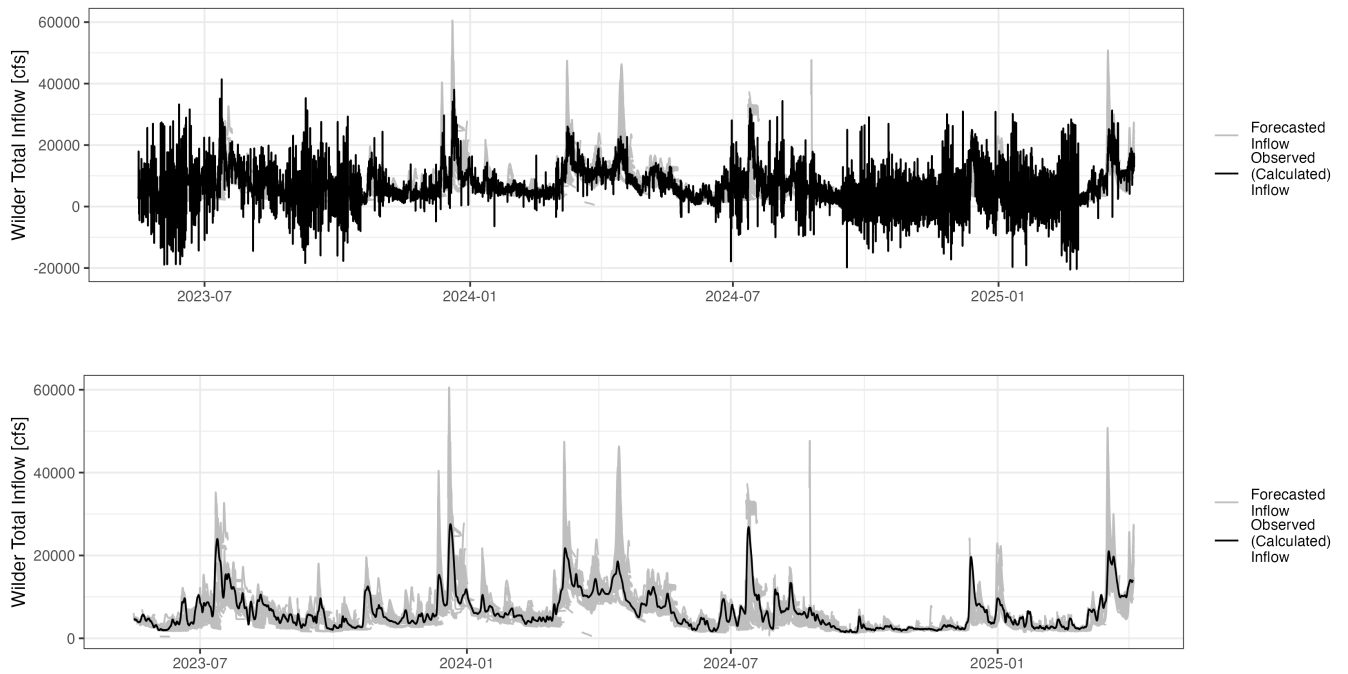


FIGURE 3 | Unsmoothed (top) and smoothed (bottom) calculated inflow at Wilder Dam. Black lines show calculated inflow derived from forebay observations and dam outflow; gray lines show forecasts. Smoothing uses two successive 24-hour moving averages, yielding a triangular kernel spanning 47 hourly observations. Flows are in cubic feet per second ($1,000 \text{ cfs} \approx 28.3 \text{ m}^3/\text{s}$).

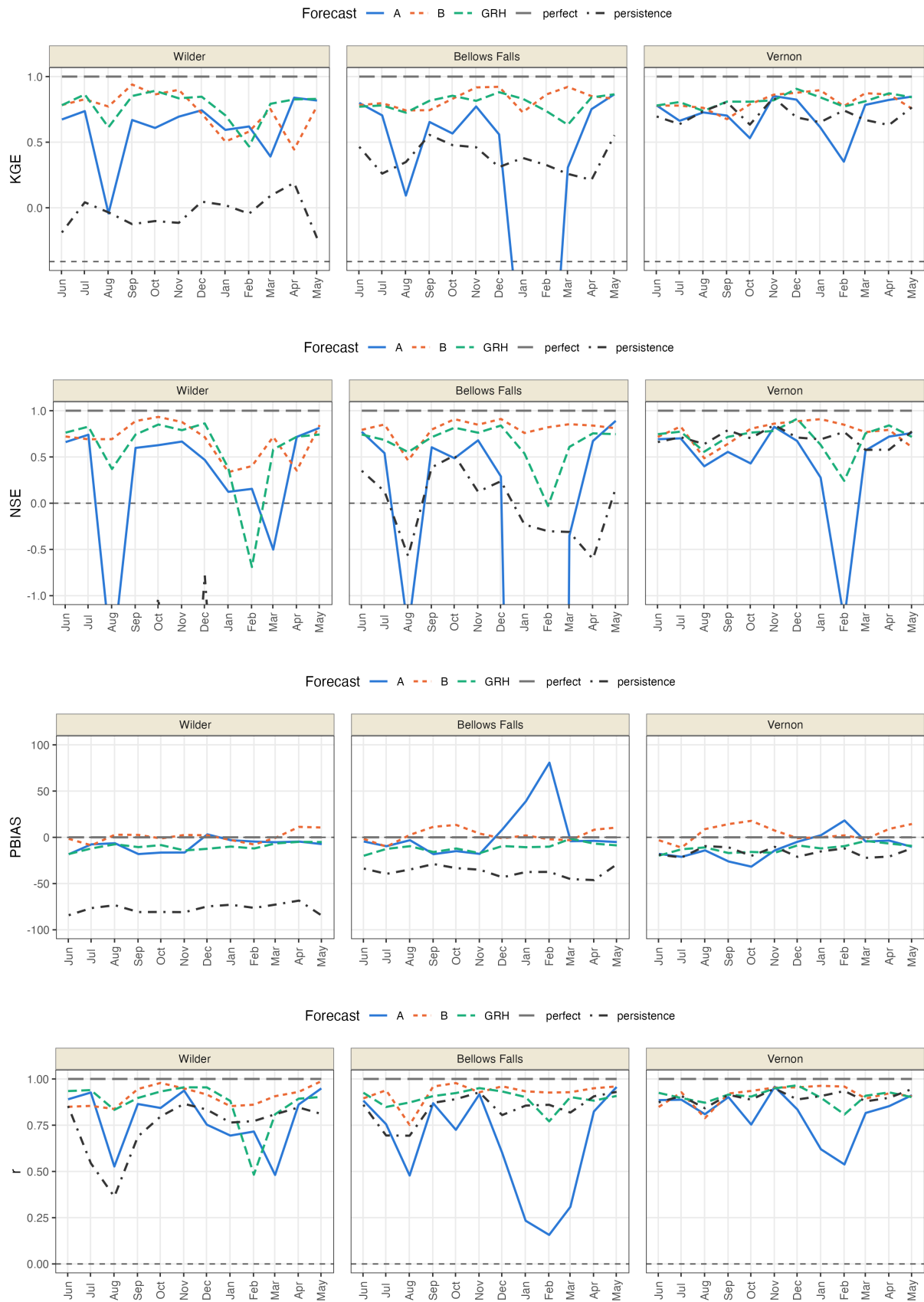


FIGURE 4 | Forecast performance metrics by month. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41 .

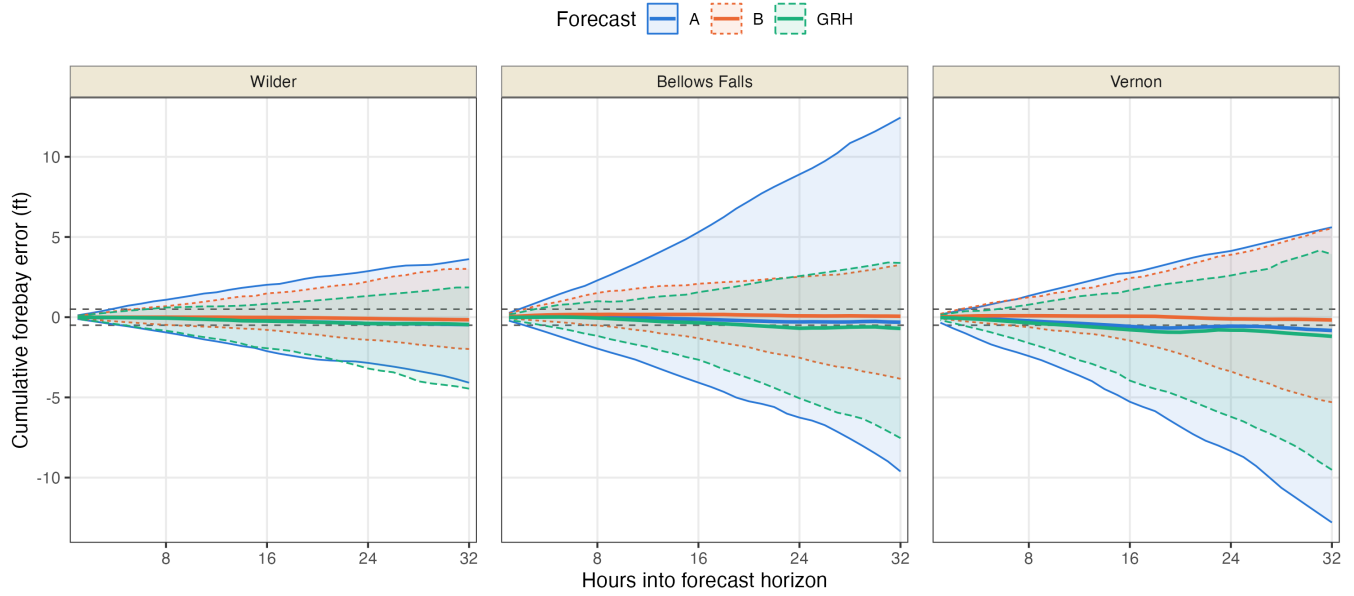


FIGURE 5 | Cumulative forebay error under an idealized pass-inflow stress test with a neutral starting position. Lines show the median across forecast issues and shaded bands span the 2.5th to 97.5th percentiles. Dashed lines mark the ± 0.5 foot (≈ 15 cm) forebay operating band. The horizontal axis counts hours from the start of each product's delivered horizon rather than absolute lead time from issue, so that every point compares an equal number of accumulated hours across products. Forecast A is issued at 04:00 but its delivered series usually begins 25 hours later; its 32 plotted hours therefore span mean lead times of 18.6 to 49.7 hours, while B and the in-house forecast begin at lead hour 1. The window ends at 32 hours, the longest horizon the in-house forecast provides.

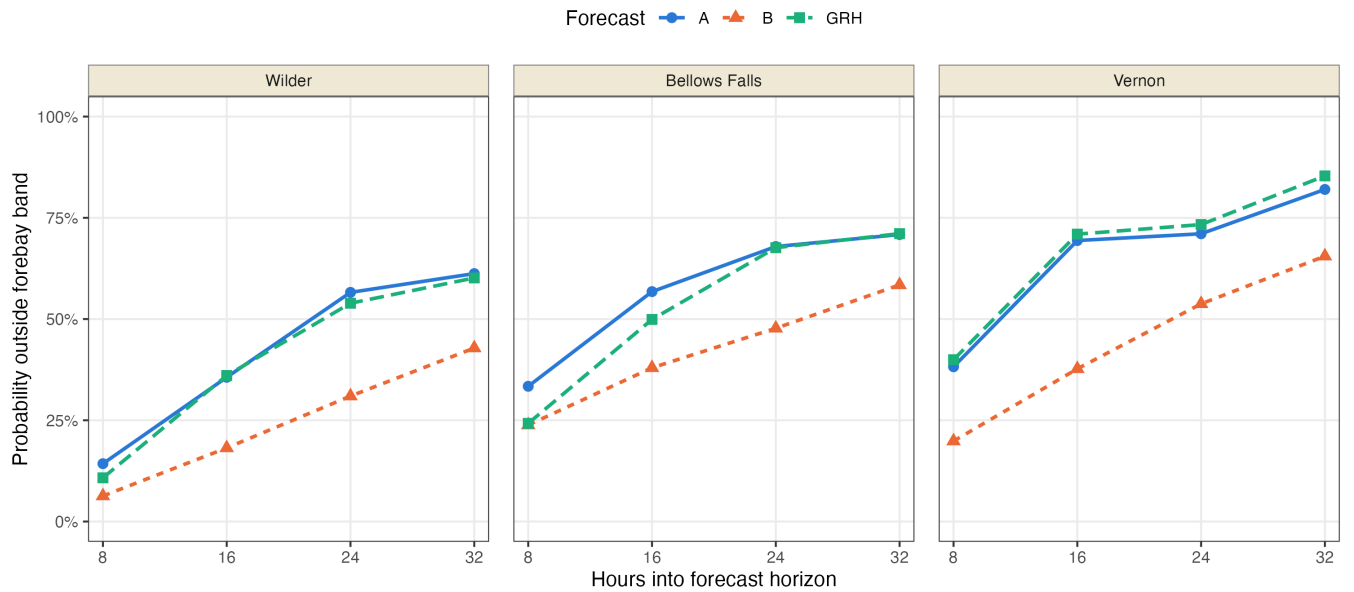


FIGURE 6 | Probability of exceeding the ± 0.5 foot (≈ 15 cm) forebay operating band under the idealized pass-inflow stress test. Each forecast issue begins from a neutral forebay position. As in Figure 5, the horizontal axis counts hours from the start of each product's delivered horizon rather than absolute lead time from issue, and the window ends at 32 hours, the longest horizon the in-house forecast provides.